

Foundation of Artificial Intelligence

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Introduction

Intelligence has been defined in many ways: the capacity to use logic, understand, have self-awareness, learn, reason, plan, be creative, think critically, and solve problems. How do we solve a problem? 1) Perceiving and learning information, 2) infer new knowledge, 3) finally making a decision. Artificial Intelligence can be used in various different fields as philosophy, mathematics, neuroscience, economics, psychology, computer engineering...

If we want a view of AI, it falls in these four categories:

- Thinking Humanly
- Thinking Rationally
- Acting Humanly
- Acting Rationally

The first one developed is Acting Humanly, in 1950 Alan Turing formalized the Turing test: “Can machines behave intelligently?”. The operational test consists in an operator talking unknownly with a human and a machine. The machine passes the test only if the operator doesn’t know who is the machine and human.

1960s “cognitive revolution”: bringing together information-processing and psychology. We want an agent able to solve problems thinking like human (behave like a human).

Thinking Rationally instead introduces logic to derive and infer. Logicians in the 19th century developed precise notations for statements about objects in the world and their relations, different from ordinary arithmetic notation, which provides only for statements about numbers. At the end acting rationally is about to have a rational behavior and so: do the right thing! The “right thing” is the one expected to maximize goal achievement, given the available information.
to be continued...